

EEG Seizure Prediction Using Logistic Regression

A comparative biomedical machine learning project for EEG seizure prediction across multiple EEG datasets using Logistic Regression, regularization techniques, PCA dimensionality reduction, and imbalance handling.

Project Overview

This project investigates EEG seizure prediction using classical machine learning techniques across three different EEG datasets:

1. UCI Epileptic Seizure Recognition Dataset
2. CHB-MIT Clinical EEG Dataset
3. Bonn EEG Dataset

The project focuses on:

- EEG feature engineering
- Logistic Regression classification
- Regularization analysis
- PCA dimensionality reduction
- SMOTE imbalance handling
- Cross-dataset generalization
- Biomedical machine learning interpretation

The study demonstrates how dataset structure, feature redundancy, and signal representation influence seizure prediction performance.

Datasets Used

1. UCI Epileptic Seizure Recognition

- Preprocessed tabular EEG dataset
- 178 EEG features
- Binary seizure classification
- Used for:
 - regularization analysis
 - PCA analysis
 - imbalance experiments

Dataset Source: <https://archive.ics.uci.edu/ml/datasets/Epileptic+Seizure+Recognition>

2. CHB-MIT Clinical EEG

- Real multichannel clinical EEG recordings
- EDF EEG format
- 23 EEG channels
- 256 Hz sampling frequency
- Statistical EEG feature engineering

Dataset Source: <https://physionet.org/content/chbmit/1.0.0/>

3. Bonn EEG Dataset

- Segmented EEG signals
- Binary seizure classification
- Compact EEG statistical representation
- Strong seizure separability

Dataset Source: https://epileptologie-bonn.de/cms/front_content.php?idcat=193&lang=3

Machine Learning Techniques

The following techniques were implemented:

Baseline Model

- Logistic Regression

Regularization

- L1 Regularization
- L2 Regularization
- Elastic Net Regularization

Imbalance Handling

- SMOTE Oversampling
- Class Weighting

Dimensionality Reduction

- PCA (Principal Component Analysis)

Additional Analysis

- Learning Curves
- Underfitting vs Overfitting
- Cross-Dataset Comparison
- Sparsity Analysis

Project Structure

```
EEG-Seizure-Prediction/
├── notebooks/
│   ├── uci_experiments.ipynb
│   ├── chbmit_experiments.ipynb
│   ├── bonn_experiments.ipynb
│   └── final_analysis.ipynb
├── figures/
│   ├── cross_dataset_performance.png
│   ├── regularization_comparison.png
│   ├── prauc_comparison.png
│   ├── sparsity_comparison.png
│   ├── pca_reduction_comparison.png
│   ├── chbmit_eeg_signal.png
│   └── bonn_eeg_signal.png
├── results/
│   ├── uci_results.csv
│   ├── chbmit_results.csv
│   ├── bonn_results.csv
│   └── cross_dataset_summary.csv
├── report/
│   ├── ieee_report.tex
│   └── ieee_report.pdf
├── README.md
└── requirements.txt
```

Key Results

Cross-Dataset Baseline Performance

Dataset	Accuracy	F1 Score	PR-AUC
UCI Epileptic Seizure	0.8161	0.1557	0.4307

Dataset	Accuracy	F1 Score	PR-AUC
CHB-MIT Clinical EEG	0.9931	0.6667	0.6131
Bonn EEG	0.9600	0.8947	0.9343

Major Findings

1. Dataset Structure Strongly Influences Performance

Performance varied significantly across datasets:

- Bonn EEG achieved strongest seizure separability
- CHB-MIT generalized well despite severe imbalance
- UCI underperformed despite high dimensionality

2. Regularization Behavior Depends on Feature Redundancy

Dataset	L1 Sparsity
UCI	21.91%
CHB-MIT	89.57%
Bonn EEG	0%

This demonstrated:

- CHB-MIT contained highly redundant engineered EEG features
- Bonn EEG features were compact and informative

3. PCA Compression Varied Across Datasets

Dataset	Original Features	Reduced Features
UCI	178	39
CHB-MIT	115	35
Bonn EEG	5	3

4. SMOTE Effectiveness Was Dataset-Dependent

- UCI: Improved F1-score
 - CHB-MIT: Limited improvement
 - Bonn EEG: Improved PR-AUC
-

Visualizations

Cross-Dataset Performance

Regularization Comparison

PR-AUC Comparison

Sparsity Comparison

PCA Feature Reduction

Technologies Used

- Python
- NumPy
- Pandas
- Scikit-learn

- Matplotlib
 - Seaborn
 - MNE
 - Imbalanced-learn
 - Jupyter Notebook
 - LaTeX
-

Installation

Clone the repository:

```
git clone https://github.com/yourusername/EEG-Seizure-Prediction.git
cd EEG-Seizure-Prediction
```

Install dependencies:

```
pip install -r requirements.txt
```

Running the Project

Run the notebooks sequentially:

1. UCI experiments
 2. CHB-MIT experiments
 3. Bonn EEG experiments
 4. Final analysis and visualization notebook
-

Future Improvements

Potential future work includes:

- Deep learning EEG classification
- CNN/LSTM seizure prediction
- Time-frequency EEG analysis
- Real-time seizure monitoring
- Explainable AI for EEG interpretation
- Patient-specific seizure prediction
- Transfer learning across EEG datasets

Research Contribution

This project demonstrates that:

- Classical machine learning models remain highly effective for EEG seizure prediction.
 - EEG feature representation strongly affects model generalization.
 - Regularization and PCA effectiveness depend heavily on dataset structure.
 - Biomedical machine learning pipelines require dataset-specific preprocessing strategies.
-

Author

Syed Mohsin Raza

Department of Data Science

Institute of Mangement Sciences Peshawar

License

This project is for educational and research purposes.